

3.2.2 Functions with Return Values

return نتیجہ کال کرنے والے پروگرام کو واپس بھیجتا ہے تاکہ وہ متغیر میں محفوظ ہو سکے یا آگے حساب میں لگے۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 11 of 17

3.2.2 Functions with Return Values

LEARN IT

A function can either **print** a result (show it on the screen) or **return** a result (send it back to the caller). A returned value can be stored in a variable, printed later, or used in a further calculation. The call is replaced by the returned value, as if you had written that value in its place.

```
def multiply(a, b):  
    return a * b  
  
result = multiply(5, 3)  
print("The answer is:", result)
```

OUTPUT

The answer is: 15

`multiply(5, 3)` evaluates to 15. That 15 is stored in `result`.

WORKED EXAMPLE

Print versus return. This is the difference examiners want.

```
def show_sum(a, b):  
    print(a + b)  
  
def give_sum(a, b):  
    return a + b  
  
x = show_sum(2, 3)  
print("stored from show_sum:", x)  
y = give_sum(2, 3)  
print("stored from give_sum:", y)  
print("double it:", y * 2)
```

OUTPUT

```
5  
stored from show_sum: None  
stored from give_sum: 5  
double it: 10
```

`show_sum` prints 5 inside the function, then returns `None` (the default when there is no `return`). You cannot double that answer later. `give_sum` hands 5 back, so `y * 2` is 10. Use `print` for messages; use `return` for results the rest of the program still needs.

REMEMBER IT

return = send back; print = show now. `return` also **exits** the function immediately. Lines after `return` in the same function do not run.

COMMON MISTAKE

Writing `print(a * b)` inside `multiply` and then `result = multiply(5, 3)` stores `None` in `result`, not 15. If you later write `result + 1` you get `TypeError`. Return the number; let the caller decide whether to print it.

اردو میں سمجھیں

`return` نتیجہ کال کرنے والے پروگرام کو واپس بھیجتا ہے تاکہ وہ متغیر میں محفوظ ہو سکے یا آگے حساب میں لگے۔ اگر `return` نہ ہو تو فنکشن `None` دیتا ہے۔ `print` صرف اسکرین پر دکھاتا ہے؛ اس سے حاصل شدہ قیمت دوبارہ استعمال نہیں ہوتی۔

CHECK YOURSELF

1. What is stored in `result` after `result = multiply(5, 3)` with `return a * b`?
2. What does a function return if it has no `return` statement?

Answers: 1. 15. 2. None .

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. return sends a value:

- a. Back to the caller
- b. To the printer always
- c. To a file always
- d. To OSI layer 1

MCQ 2. A function with only print:

- a. Shows something but gives None to the caller
- b. Returns the text automatically
- c. Returns 0
- d. Cannot run

MCQ 3. `x = f()` if `f` has no return:

- a. `x` is None
- b. `x` is 0
- c. Error always
- d. `x` is `f`

MCQ 4. return also:

- a. Exits the function immediately
- b. Loops forever
- c. Defines a class
- d. Opens a socket

MCQ 5. You can return:

- a. Any object (number, str, tuple, list, dict)
- b. Only ints
- c. Only True
- d. Only files

MCQ 6. `print(f())` prints the:

- a. Returned value (or None)
- b. Source code
- c. The name `f`
- d. A pie

Extra short questions

1. Why return rather than print inside?
2. Show both styles for doubling 5.
3. Returning two values.
4. What is None?

Extra long question

1. Write `mean(a,b,c)` that returns a float. Show a test print and a logic error version that prints but returns None. Explain the bug in a unit-test style.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. The caller can store, test, or reuse the value.
2. print inside vs return 10 and print in main.
3. return a,b → a tuple, unpack.
4. The default return if you fall off the end.

Long

1. Correct return (a+b+c)/3. Buggy print only → assert mean(1,2,3)==2 fails with None.
Teaching point of return.

Next lecture: 3.2.3 Activity: Arithmetic Calculator Using Functions